

EDUCATION

Northeastern University, Boston

Sept 2025 – Dec 2027

MS, Computer Science

Relevant Coursework: Fundamentals of Artificial Intelligence, Algorithms, Database Management Systems, PDP

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, Java, C++, SQL, Bash
- **Frameworks & Development:** React, Next.js, Angular, Node.js, Express.js, FastAPI, Spring Boot, REST APIs, MVC Architecture
- **Machine Learning & AI:** LLM Applications, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), AI Agents, Grounded LLM Outputs, Prompt Engineering, Vector Embeddings, ChromaDB, Llama 3.2, Claude API, PyTorch, NLP, Anomaly Detection, Time Series (ARIMA/LSTM)
- **Cloud, Infra & Databases:** AWS (S3, RDS), Vercel, PostgreSQL, Prisma, MongoDB, MySQL, ChromaDB, Docker, CI/CD (GitHub Actions), Linux, Git
- **Computer Science:** Data Structures & Algorithms, Object-Oriented Programming, Distributed Systems, Concurrency, System Design

WORK EXPERIENCE

VidarthiMitra.org | Web Developer

Jun 2024 – Dec 2024

- Optimized MySQL queries and added server side caching, cutting page load times 30% across a platform serving 1,000+ users
- Built REST APIs in Node.js and reusable Angular components in an Agile team, shipping features across multiple sprints

Texas A&M University, Texarkana | Research Intern

Jun 2024 – July 2024

- Developed ARIMA and LSTM machine learning models in Python to forecast electricity consumption trends, benchmarking predictive accuracy across multiple datasets and winning 1st Place at the university research symposium
- Designed and optimized large-scale data processing pipelines for historical energy datasets, automating experimentation workflows and accelerating model evaluation
- Collaborated with faculty researchers to prototype machine learning solutions, analyze model performance, and translate research objectives into scalable data-driven insights

Outlier.ai | AI Contractor (Freelance)

Jan 2024 – Jun 2024

- Curated and validated large scale datasets for production LLM training and evaluation, improving quality through verification, prompt refinement, and error detection
- Evaluated AI outputs across diverse tasks, identifying failure patterns and edge cases to improve model reliability and reasoning

PROJECTS

FinPulse — AI Spending Insights Platform | TypeScript, Next.js, Python, FastAPI, MCP, Claude, PostgreSQL | [Link](#)

- Built a read only Model Context Protocol (MCP) server exposing 5 tools that let AI agents such as Claude Desktop query spending, trends, and anomalies directly against a live PostgreSQL database
- Engineered grounded natural language insights that assemble real aggregate figures from Postgres into the prompt and constrain Claude to answer only from those numbers, returning the exact data each answer was grounded on for full auditability
- Implemented two anomaly detection strategies in a Python and FastAPI service (robust median/MAD statistical scoring versus ARIMA forecasting) behind a shared interface, and benchmarked them against planted anomalies, with the statistical detector reaching 1.00 precision and recall

Personal Email RAG System | Python, LLaMA 3.2, ChromaDB, Gmail API, OAuth2 | [Link](#)

- Built an end-to-end AI retrieval system using Python, Llama 3.2, ChromaDB, and vector embeddings, enabling low-latency semantic search and automated knowledge retrieval across 1,000+ indexed emails
- Designed agent-style retrieval workflows integrating LLM reasoning, embedding-based search, and contextual response generation to automate information discovery and question answering
- Architected scalable retrieval infrastructure supporting autonomous document processing, workflow automation, and concurrent user queries while maintaining secure data isolation

IT Service Management System | Angular, Node.js, PostgreSQL | [Link](#)

- Designed and developed a full-stack enterprise platform using Angular, Node.js, and PostgreSQL to manage incidents, assets, and change requests
- Built secure REST APIs with JWT authentication and role-based access control supporting multiple business workflows
- Implemented modular service-layer architecture to improve maintainability, scalability, and performance across 10+ application modules